

Sleeves & Openings Standards Manual

Organized Version

All rule wording has been preserved. Formatting and section organization have been improved.

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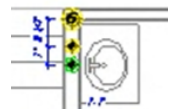
General Rule: Always use the Owner's plan Instead of Approved plans.

General Rule: Always check and confirm that the file you are using is the latest version before starting any work.

General Rule: never put a sleeve by the edge of wall since there is no possible way for the sheetrock studs

General Rule — Always Verify:

- Shower drain type
- Toilet type — especially if it's **wall-hung**
- Medicine cabinets
- Niches



Exhaust Openings

1. Set the View Range to Level Above 0.
2. Set the View Range Bottom to Associated Level.
3. Exhaust risers usually start smaller on lower floors and increase in size as they rise toward the roof fan.
4. Most exhaust risers terminate at roof-mounted fans.
5. There is also many times where there is a gooseneck which is just an exhaust
6. Kitchens, toilets, and trash rooms typically have exhaust systems.
7. Some exhaust systems discharge through walls, but most continue to the roof.
8. Start on the highest floor that contains apartments.
9. Open the Mechanical Engineer's plan PDF.
10. Arrange the PDF to view both the floor plan and riser diagram.
11. Load the Regular Opening family into the project.
12. Press M+E to open the Opening Tool.
13. Place an opening for every exhaust riser shown on the floor plan.
14. Name each opening according to project requirements.
15. Size openings using the floor plan dimensions.
16. Add 2 inches of clearance on each side of the duct.
17. If the riser diagram and floor plan conflict, always follow the floor plan.
18. Verify all openings have the correct size and name.
19. Relocate openings to save apartment space when possible.
20. Avoid placing openings in the middle of rooms.
21. Prefer walls, closets, shafts, chases, and other dead spaces.
22. Avoid offsets whenever possible.
23. Keep risers straight and aligned from floor to floor.
24. Keep risers the same size whenever possible.

25. If a riser must offset, use the size shown on the Mechanical Engineer's plan to decrease the size.
26. Keep openings at least 1 foot away from columns whenever possible.
27. Do not place openings in concrete shear walls.
28. Do not place openings in structural beams.
29. Always consider the entire riser path through the building.
30. Review the riser diagram carefully for tap-outs.
31. A tap-out is typically labeled KX or TX Tap-Out.
32. Some risers terminate before reaching the first floor.
33. Some risers terminate on intermediate floors.
34. Always verify where each riser starts and terminates.

Garbage Chute Rules

- 34. Garbage chute openings are always 28.5" x 28.5".
- 35. Do not increase or decrease garbage chute opening sizes.
- 36. Garbage chutes carry trash from the building to the trash collection area.
- 37. Garbage chutes remain in the same location throughout the building.
- 38. Garbage chutes usually terminate at the roof.
- 39. Some garbage chutes terminate at a bulkhead.
- 40. There is typically an exhaust fan at the top of the garbage chute.
- 41. Check the architectural background to determine where the chute terminates.
- 42. Garbage chutes must remain in a straight line.
- 43. Garbage chutes do not offset.
- 44. Do not add clearance to garbage chute openings at the roof.

Please note: Not all buildings are required by code to have a garbage chute.

Lower Floors

45. Copy all openings from the highest apartment floor to the floor below.
46. Use Select All Instances Visible in View.
47. Continue copying openings downward floor by floor.
48. Keep openings in the same location whenever possible.
49. Keep openings the same size whenever possible.
50. Avoid unnecessary offsets between floors.
51. Verify where each riser terminates as you move down the building.

Rooftop Instructions

52. After completing the building, copy all openings from the highest apartment floor to the roof.
53. Use Select All Instances Visible in View and past to current view.
54. ERV openings should always be grouped together.
55. Maintain exactly 2 feet between ERV openings.
56. Increase roof exhaust openings by 4 inches total.
57. Verify roof opening sizes using the riser diagram.
58. Maintain at least 1 foot of clearance from walls and curbs.
59. Maintain at least 2 feet between openings.
60. Do not place openings in roof fire paths.
61. Keep openings as close as possible to their riser locations.
62. Avoid unnecessary offsets at the roof.
63. Coordinate openings that continue to bulkheads.
64. Place bulkhead-related openings in indoor roof areas when required.
65. Keep roof openings organized and aligned.
66. Do not randomly scatter openings across the roof.
67. If a riser terminates on a setback, apply the same rules used for roof openings.

Dryer Exhaust Vents

68. Model dryer exhaust vents as circular openings inside a Regular Opening box.
69. Press DL to draw the dryer exhaust opening.
70. Select the Circle option.
71. Enter a radius of 2 inches.
72. A 2-inch radius creates a 4-inch round dryer exhaust.
73. Name the opening DE for Dryer Exhaust.
74. Place the circle inside a mechanical opening box.
75. Dryer exhaust ducts are typically 4 inches round.
76. Carefully review all Mechanical plans for dryer exhaust vents.
77. Continue dryer exhaust vents to the roof when required by the Mechanical Engineer.
78. Create a 6" x 6" opening for each 4" dryer exhaust at the roof.
79. Maintain at least 8 inches between dryer exhaust openings.
80. Align dryer exhaust openings in straight rows whenever possible.
81. Keep dryer exhaust openings organized on the roof.
82. Do not randomly scatter dryer exhaust openings.

Stair Pressurization & Motorized Dampers

- 83. Review Mechanical plans for motorized dampers above stairways.
- 84. Add 2 inches of clearance on each side of motorized damper openings.
- 85. If a motorized damper penetrates a bulkhead, coordinate the opening accordingly.
- 86. Minimize offsets whenever possible.
- 87. This motorized damper contains the same rules as all the roof exhaustions

REFRIGERATION LINES – QUICK RULES

1. Check the AC system type.
2. If PTAC units are used, no refrigeration line coordination is required.
3. Check the condenser model online.
4. Determine whether the system is:
 - Shared refrigeration pair (VRF/VRV type)
 - Individual refrigeration pairs (traditional split system)
5. For VRF/VRV systems:
 - Multiple indoor units may share the same refrigeration pair.
 - Do not count 2 lines for every indoor unit.
 - Most projects use single-port VRV systems.
6. For traditional split systems:
 - Every indoor unit requires:
 - (1) Suction Line
 - (1) Liquid Line
 - Count 2 refrigeration lines per indoor unit.
7. Use Pipe Reference Opening and verify Down Height = 0.
8. Group apartments by stack. Try to group all A apartments together, all B apartments together, etc.
9. Calculate refrigeration lines based on the actual system design.
10. Verify the location of every condenser.
11. Condensers may be located on:
 - Main Roof
 - Setback Roof
 - Terrace Roof
 - First Floor Roof
 - Lower Roof

12. Do not assume all condensers are located on the highest roof.

13. If refrigeration lines already come from below, do not create additional penetrations above.

14. Many condensers are stacked directly above each other. Coordinate stacks whenever possible.

15. Many projects use one condenser for all corridor units.

16. Example:

- 10 corridor units
- 1 condenser
- Only 2 refrigeration lines required

17. Avoid placing openings in the middle of apartments.

18. Use closets, shafts, mechanical rooms, and utility spaces whenever possible.

19. Example:

- 10 floors
- 10 apartments per floor
- Floors 1–5 condensers located on the 2nd-floor roof
- Floors 6–10 condensers located on the main roof

In this case, the refrigeration lines serving Floors 1–5 come from below. Therefore, an additional penetration at Floor 6 is not required.

20. On the roof, align refrigeration line openings with the condenser layout.

21. If condensers are stacked in groups, place openings strategically between the condenser groups.

22. Example:

- 6 condensers in a row
- Create one opening serving the first 3 condensers
- Create a second opening serving the other 3 condensers

23. On the roof Allow additional 3 inch space in the width and the length for the refrigeration line openings for electrical conduits serving the condensers.

24. If refrigeration lines must continue through a bulkhead, determine the route early and locate the penetration in an indoor location whenever possible.

25. Verify that you covered every condenser in the entire building

ELECTRICAL OPENING RULES

1. The blue box represents the Electrical Opening.
2. Start by locating the Electrical Room in the cellar/lowest floor.
3. Place the Electrical Opening as close as possible to the Electrical Room while maintaining a clear path through the building.
4. Electrical conduits originate from the lowest floor and rise up through the building.
5. Provide one conduit circle for each apartment.
6. Provide one additional conduit circle for the roof.
7. Example: A 10-story building with 10 apartments per floor requires 101 circles at the lowest floor (100 apartments + 1 roof).
8. Draw all conduit circles using a detailed line.
9. Space conduit circles 0.75" center-to-center.
10. Arrange the circles to keep the conduits as straight as possible and avoid offsets.
11. If there is no offset, keep the same 101 circles all the way up the building.
12. Only reduce the number of circles and the opening size when an offset is required.
13. Example: If an offset occurs above the 5th floor, remove the circles serving Floors 1-5 (50 apartments). You will now need 51 circles (50 apartments remaining on Floors 6-10 + 1 roof).
14. Recalculate the opening size based on the remaining circles after each offset.
15. Continue this process until reaching the roof.
16. At the roof, only one conduit remains. Provide one 2" ELECTRIC sleeve.
17. Try to keep the roof sleeve indoors whenever possible. Find a location inside the building envelope, mechanical room, stair enclosure, chase, or wall cavity.
18. Label all sleeves and openings as ELECTRIC.

Please note: If there is no electrical room, assume electrical services come from the mechanical room and place electrical openings as close to the mechanical room as possible.

Storm Sleeves – Quick Rules

1. Open the Plumbing Floor Plans and Storm Riser Diagram.
2. Start at the highest roof or terrace level and work down.
3. Identify all storm drains (RD, AD, garage drains, entrance drains, etc.).
4. Follow the riser diagram and determine which drains connect to each riser.
5. Plan the entire riser path first—from the highest drain to the underground storm system.
6. Place the riser sleeve as close to the area drain or roof drain as possible to avoid offsets and ceiling drops.
7. Keep risers near walls, shafts, closets, or other concealed locations. Avoid the middle of rooms.
8. Area Drains (AD) are always 10".
9. When two Area Drains (AD) are shown together, maintain 2'-0" center-to-center spacing.
10. On detail decks, place the Area Drain 1'-0" from the slab edge unless otherwise noted.
11. Sleeve size = Pipe size + 2".
12. Verify all drains are connected and all risers continue to the underground storm system.

Note: Even if the engineer shows only **one Area Drain (AD)**, always place **two 10" Area Drain sleeves**. One drain is the **primary drain**, and the second is the **overflow drain**. Maintain **2'-0" center-to-center** spacing between the two sleeves.

Final Check

- ✓ All drains accounted for
- ✓ 10" Area Drains used
- ✓ 2'-0" center-to-center spacing maintained
- ✓ Detail deck drain located 1'-0" from slab edge
- ✓ Minimal offsets
- ✓ No unnecessary ceiling drops
- ✓ Sleeve size correct
- ✓ Risers coordinated to the underground storm system

Condensate Sleeves – Quick Rules

1. Open the mechanical floor plans. First, verify with me whether the building requires condensate risers. PTAC units may also require
2. Start at the highest floor and identify all AC units that produce condensate.
3. Place the condensate riser sleeve as close to the unit as possible, preferably in the nearest wall, chase, or directly behind the unit.
4. If two units are back-to-back, use one shared condensate riser sleeve whenever possible.
5. All condensate sleeves are 3".
6. Copy the sleeve down through all floors and verify every condensate-producing unit is covered.
7. Continue the riser to the lowest floor and connect it to the underground drainage system.

Final Check

- ✓ Building requires condensate risers
- ✓ All units covered
- ✓ 3" sleeve used throughout
- ✓ Shared risers used where possible
- ✓ Minimal offsets
- ✓ Connected to underground drainage system

Sprinkler Standpipe Sleeves – Quick Rules

Open the Fire Protection Plans and Riser Diagram.

Identify all standpipes shown on the riser diagram and verify their sizes.

Most buildings have two stairwells:

One Standpipe Only.

One Combined Standpipe/Sprinkler Riser.

The Standpipe Only riser is for firefighter hose connections and typically has hose valves on every floor.

The Combined Standpipe/Sprinkler Riser supplies the floor sprinkler system and also serves as a standpipe.

Place sleeves in the stairwell whenever possible and keep the riser in the same location from top to bottom.

Maintain 5½" minimum clearance from the center of the pipe to any wall.

When a Combined Riser and Standpipe are next to each other, maintain 8" minimum center-to-center spacing between pipes.

Sleeve size = Pipe size + 2".

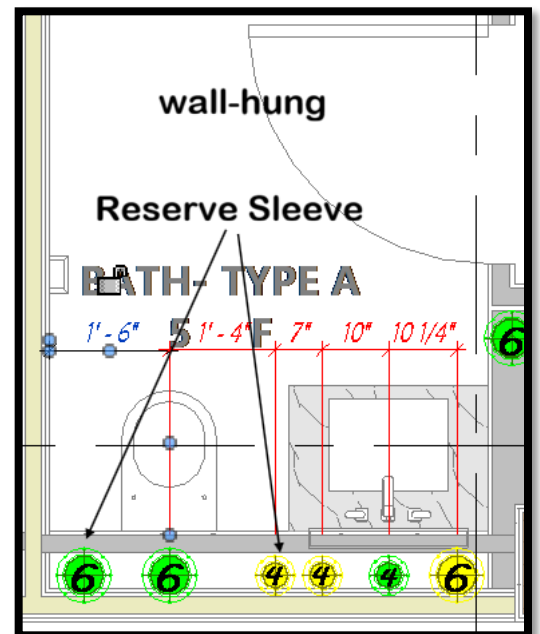
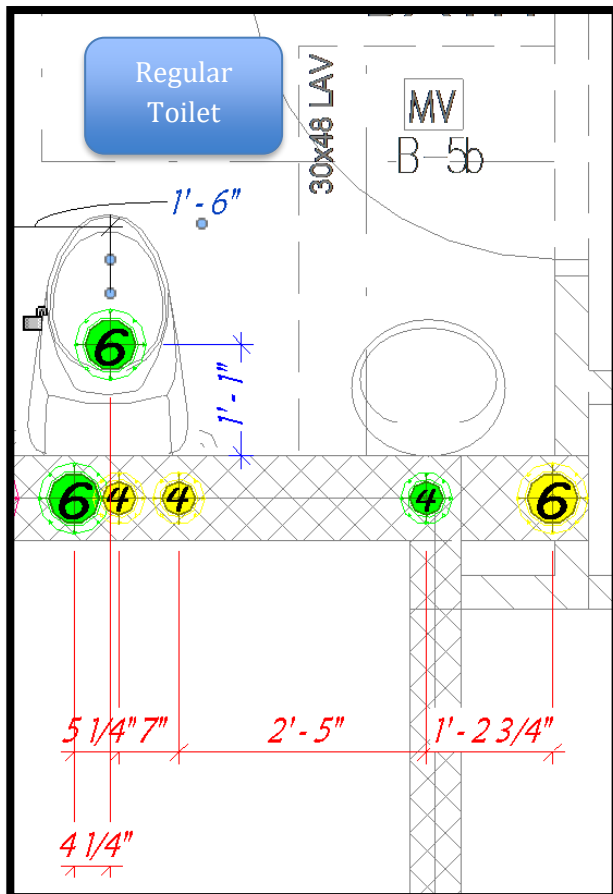
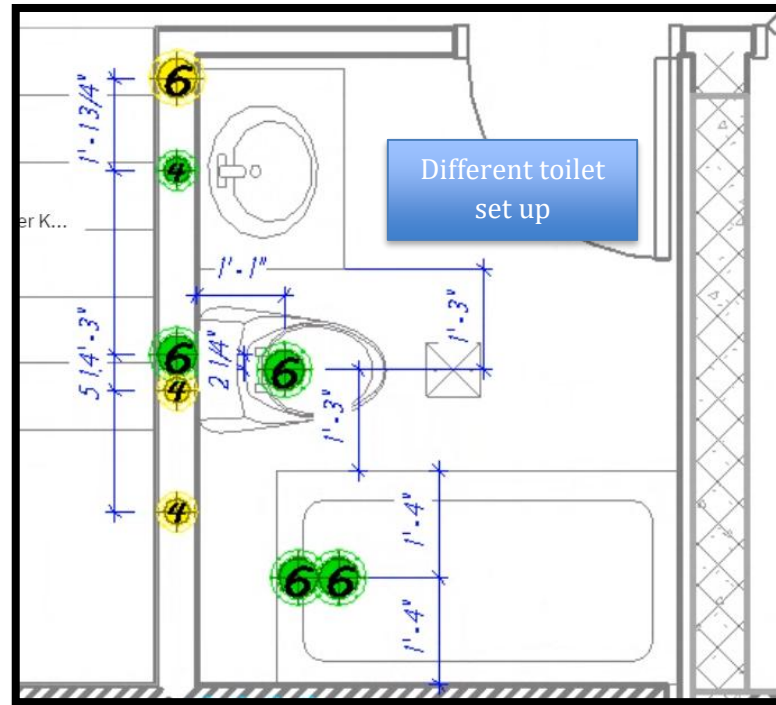
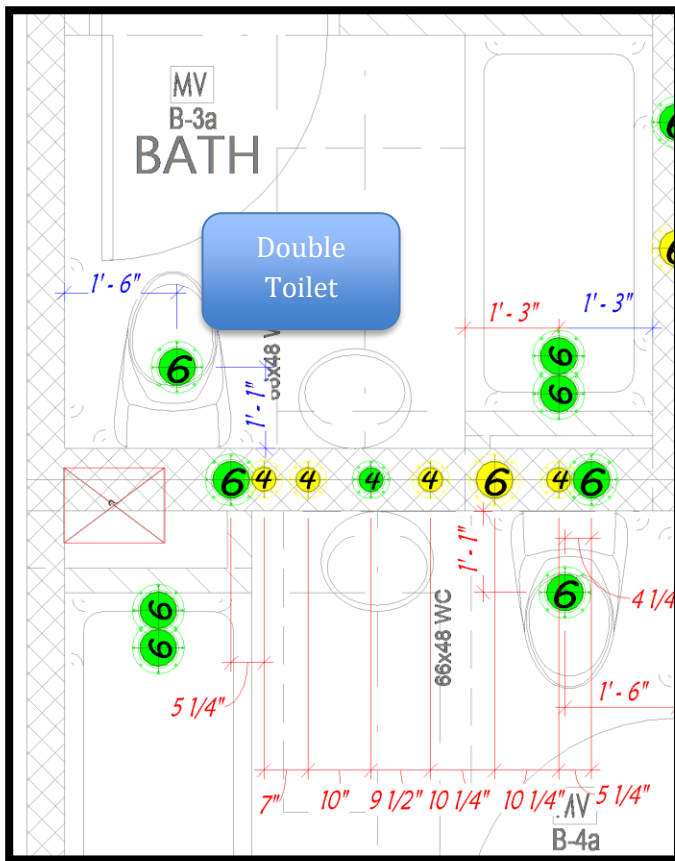
Copy the sleeves through all floors and verify they reach the lowest level.

Check for additional standpipes beyond the two stairwells. Large buildings may have multiple standpipes.

Final Check

- ✓ All standpipes identified from riser diagram
- ✓ 5½" minimum centerline-to-wall clearance maintained
- ✓ 8" minimum center-to-center spacing maintained
- ✓ Sleeve size correct
- ✓ Risers aligned through all floors
- ✓ Additional standpipes accounted for
- ✓ Risers continue to the lowest level

Toilet sleeves



Bathtub sleeves

Before starting, always ask how the **bathtub sleeve** should be done:

- **Two 6" sleeves, or**
- **One 10" sleeve.**
- **Death center**